

Curve, Freeze, Thaw: HPO with FT-PFN

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Team auto-auto

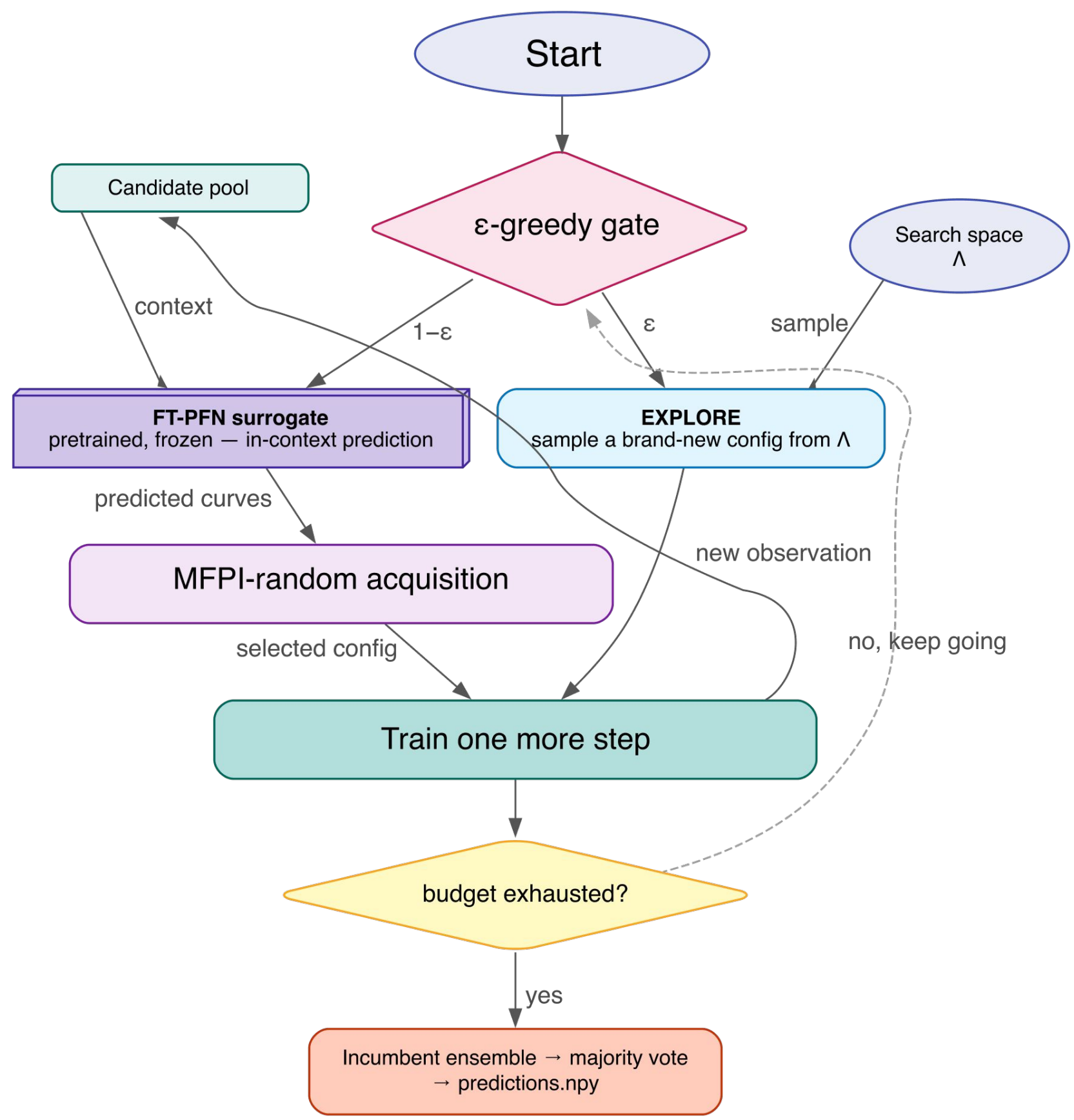
Modality 3/3

In a Nutshell

- Standard optimization wastes budget by training every candidate to completion, *even the obvious dead ends*.
- Our Approach: **Multi-fidelity, freeze-thaw Bayesian optimization**. Configurations are trained *partially* and in *parallel*, free to pause or resume based on its learning curve's prediction.
- Reaches target accuracy using **~60%** less wallclock time than random search across 4 text-classification datasets

The Approach

- At each step, the optimizer picks an action $a \in \{\text{start a new config } \theta, \text{ thaw a partially-trained one}\}$, maximizing validation performance under a total-step budget B .
- Explore or exploit?** With decaying probability ϵ , sample a brand-new configuration. Otherwise, exploit.
- Exploit.** FT-PFN, a pretrained, meta-learned surrogate, scores every viable candidate from its partial learning curve.
- MFPI-random acquisition.** FT-PFN estimates each candidate's probability of beating a random target within a random number of further steps; the highest-scoring candidate is selected.
- Warm-started embeddings:** BiLSTM embeddings train from scratch but are seeded from a pretrained token embeddings, SVD-projected down to the tuned embedding dimension.



Safety in Numbers

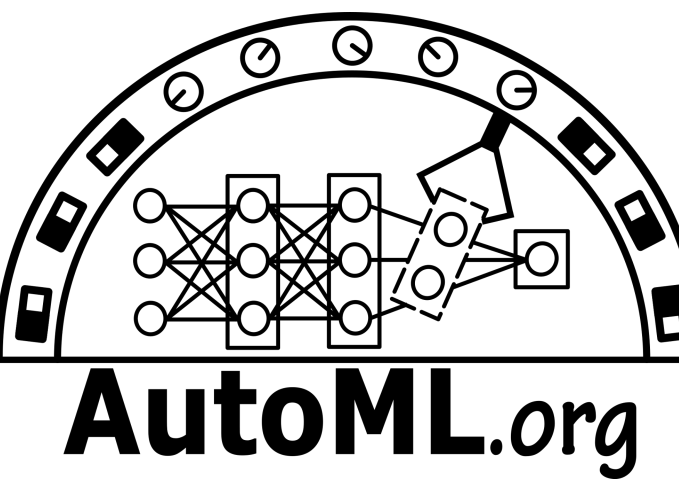
- Top-k selection.** Take the k best-performing configs by validation accuracy.
- Quality threshold.** Only configs within a small margin of the best score qualify to keep clearly weak models out of the ensemble.
- Majority vote.** Final prediction is the ensemble's majority vote across the selected models.

- ☐ Week 1
- ☐ Week 2
- ☒ Week 3
- ☒ Week 4
- ☒ Week 5
- ☐ Week 6
- ☐ Week 7
- ☐ Week 8
- ☐ Week 9
- ☐ Week 10
- ☒ Bonus
- ☒ Literature

Resources Used

- For development:
- 1 × Apple Silicon GPU (M2 Max integrated GPU)
 - Total compute estimate: ~1000 CPU-h
- For AutoML:
- Kaggle with T4 GPU
 - 10h
- Workforce:
- 2 full week on average

Number of queries for test score generation: 2



Search Space & Fidelity

Fidelity dimension: training epochs, as it is simple to extend incrementally, the natural fit for learning-curve methods.

Configuration space: Capped at FT-PFN's architectural limit of 10 encoded dimensions. The surrogate could be retrained for higher dimensions per the original paper.

BiLSTM Model: the preferred model used for predicting the training curves, due to its strong performance-to-cost ratio, making it cheap enough to train hundreds of configurations per dataset.

Hyperparameter	Range/Choices	Default	Why?
hidden_dim	32 - 256 log	128	capacity vs overfitting/compute
seq_num_layers	1-3	1	depth – lightweight arch. search
seq_embed_dim	32 - 512 log	128	representation capacity
dropout	0.0-0.5	0.2	regularization
learning_rate	1e-4 - 1e-2 (log)	1e-3	
weight_decay	1e-6 - 1e-2 (log)	1e-4	regularization
optimizer	{adam, adamw, sgd}	adamw	algorithm selection
scheduler	{steplr, cosine, exp., reduce}	cosine	LR-schedule selection (alg. selection)
pre. model name	{distilbert, bert, google/bert, xtremedistil}	distilbert	Pretrained embedding
batch_size	32 - 512 log	64	compute/stat. efficiency
max_seq_length	64 - 256 log	128	

Empirical Results

